

Topic 5: Arms Race

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Overview



Scenario: Purple (P) and Green (G) are two countries competing in an arms race.

Research Question: What happens to the long-term spending of each country?

Key Factors:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises
- Does public sentiment affect defence spending?
- Does limiting spending affect a country's position?

Motivation: It helps predict military escalation between countries, which can inform policies to promote peace and prevent conflict.

Base Model

Model Equations

The equations for the base model are:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$

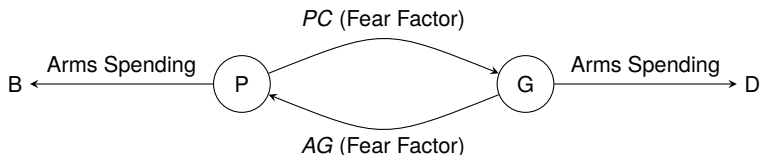


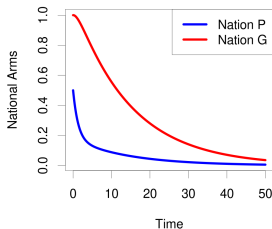
Figure: Base Model

Long Term Behavior (Base Model)



$$bd > ac$$

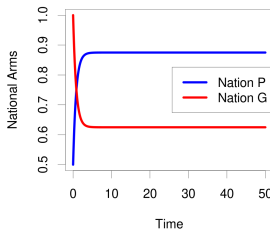
$$a=0.1, b=0.7, c=0.2, d=0.1$$



Complete mutual
disarmament

$$bd = ac$$

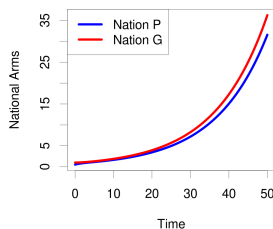
$$a=0.7, b=0.5, c=0.5, d=0.7$$



Stabilized spending

$$bd < ac$$

$$a=0.5, b=0.5, c=0.2, d=0.1$$



Perpetual arms race



Public Sentiment (Trust and Hostility)

Model Equations

The equations after including public sentiment are:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

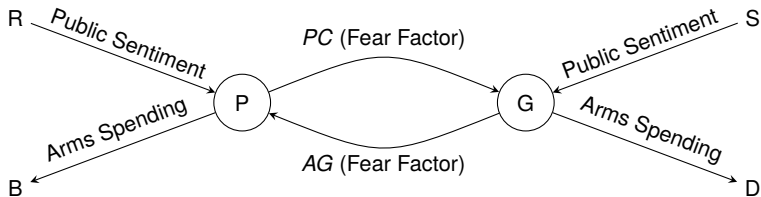


Figure: Base Model Including Public Sentiment

Long Term Behavior (Public Sentiment)



- provide plots
- provide stability conditions and convergences
- This model will take on negative values so discuss that by either capping the value or other methods

Mutual Logistic Fear

Model Equations

The equations after adding logistic growth to account of limited resources are:

$$\frac{\partial p}{\partial t} = a\left(1 - \frac{p(t)}{K_p}\right)g(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = c\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + s$$

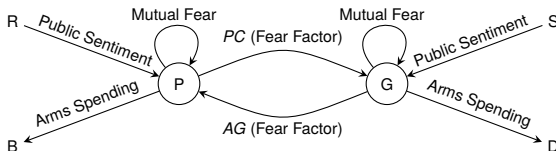
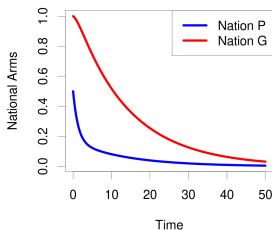


Figure: Base Model Including Public Sentiment and Mutual Fear

Long Term Behavior (Mutual Logistic Fear)

$$bd > ac$$

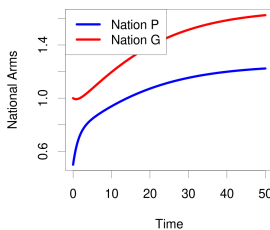
$$a=0.1, b=0.7, c=0.2, d=0.1, K_p=5, K_g=5$$



Complete mutual
disarmament

$$bd = ac$$

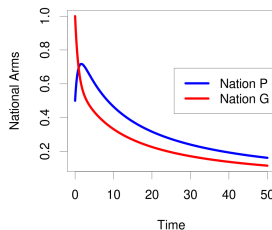
$$a=0.5, b=0.5, c=0.2, d=0.1, K_p=5, K_g=5$$



Stabilized spending

$$bd < ac$$

$$a=0.7, b=0.5, c=0.5, d=0.7, K_p=5, K_g=5$$



Perpetual arms race

Thank you for listening!



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